RESCALED FAMILIES AND SUMS OF TWO SQUARES IN THREE-TERM PATTERNS AND ALONG $n^3 + a^3$

CLAUDE (ANTHROPIC)

ABSTRACT. For a fixed integer $a \geq 1$ let $C_a(x)$ be the number of $n \leq x$ such that $n^3 + a^3$ is a sum of two squares, and for fixed integers $0 < a < b$ let $S_{a,b}(x)$ be the number of $n \leq x$ such that n, $n+a$ and $n+b$ are all sums of two squares. We prove that $C_a(x) \gg x^{1/2+\delta}$ and $S_{a,b}(x) \gg x^{1/2+\delta}$ for every

$$\delta < \frac{1-2\theta}{8-8\theta},$$

where θ is any exponent towards Selberg's eigenvalue conjecture for the groups $\Gamma_0(q)$. The Kim–Sarnak bound $\theta = 7/64$ gives every $\delta < 25/228 = 0.1096\ldots$, and Selberg's conjecture would give every $\delta < 1/8$. For three-term patterns this improves the exponent $(1-2\theta)/(10-12\theta)$ of the earlier work [4], which is $25/278 = 0.0899\ldots$ for $\theta = 7/64$, and it exceeds the value $1/10$ that [4] obtains under Selberg's conjecture.

The method of [4] makes two of the three numbers sums of two squares by an identity attached to a parameter d, and counts the values of a quadratic polynomial whose roots are governed by Heegner points of discriminant $\asymp d^2$ on $\Gamma_0(4Ed)$. Its Type I estimate loses a factor that depends on the discriminant but not on the level. We introduce a second parameter r, a prime $\equiv 1 \pmod 4$. The rescaled families have level rd and discriminant $\asymp (r+d)^2$, so for $r \asymp d$ the discriminant is of the size of the level instead of its square. For $n^3 + a^3 = m\,(m^2 - 3am + 3a^2)$ with $m = n + a$, the analogous families come from the identity $(2\ell)^2 + (2rm - 3ar - D)^2 = 4r^2(m^2 - 3am + 3a^2)$, which holds whenever $rDm = \ell^2 + \left(\frac{D+3ar}{2}\right)^2 - 3a^2r^2$. The analytic input is the Type I estimate of [4], which we extend to levels that are products of two primes.

1. INTRODUCTION

Let $\mathcal{S} = \{u^2 + v^2 : \ u, v \in \mathbb{Z}\}$ and $r(n) = \#\{(u,v) \in \mathbb{Z}^2 : \ u^2 + v^2 = n\}$. For fixed integers $a \geq 1$, and $0 < a < b$ respectively, put

$$C_a(x) = \#\{1 \leq n \leq x : \ n^3 + a^3 \in \mathcal{S}\}, \qquad S_{a,b}(x) = \#\{1 \leq n \leq x : \ n,\ n+a,\ n+b \in \mathcal{S}\}.$$

Writing $m = n + a$ and $G_a(m) = m^2 - 3am + 3a^2$, we have $G_a(n+a) = n^2 - an + a^2$ and

$$n^3 + a^3 = m\, G_a(m). \tag{1.1}$$

Both factors impose a condition of sieve dimension $\frac{1}{2}$, so one expects $C_a(x) \asymp x/\log x$, and Selberg's sieve gives an upper bound of this order. For three-term patterns the conjecture of Freiberg, Kurlberg and Rosenzweig [3, Conjecture 1.1] predicts $S_{a,b}(x) \sim \mathfrak{S}\, C^3 x (\log x)^{-3/2}$, and upper bounds of this order are known [1, 6]. Lower bounds are much weaker. One-parameter constructions give about $x^{1/2}$ elements up to x. In [4] it is shown that $S_{a,b}(x) \gg x^{1/2+\delta}$ for every $\delta < (1-2\theta)/(10-12\theta)$.

As in [4], for a congruence subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ let $\theta(\Gamma) = \max\{0, \mathrm{Re}\,\sqrt{1/4 - \lambda_1(\Gamma)}\}$, where $\lambda_1(\Gamma)$ is the smallest positive eigenvalue of the Laplacian on $\Gamma\backslash\mathbb{H}$. We call $\theta \in [0, \frac{1}{2})$ *admissible* if $\theta(\Gamma_0(q)) \leq \theta$ for every $q \geq 1$. By Selberg [8] $\theta = 1/4$ is admissible, by Kim and Sarnak [5, Appendix 2] $\theta = 7/64$ is admissible, and Selberg's eigenvalue conjecture is that $\theta = 0$ is admissible.

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Put

$$\delta_\theta^* = \frac{1-2\theta}{8-8\theta}, \qquad \delta_\theta^{**} = \frac{1-2\theta}{8-4\theta}.$$

Theorem 1.1. *Let $a \geq 1$ be a fixed integer and let θ be admissible. For every $\delta < \delta_\theta^*$ we have $C_a(x) \gg_{a,\delta} x^{1/2+\delta}$ for $x \geq x_0(a,\delta)$.*

Theorem 1.2. *Let $0 < a < b$ be fixed integers and let θ be admissible. For every $\delta < \delta_\theta^*$ we have $S_{a,b}(x) \gg_{a,b,\delta} x^{1/2+\delta}$ for $x \geq x_0(a,b,\delta)$.*

The implied constants in Theorems 1.1 and 1.2 are ineffective, because Siegel's theorem is used (Remark 7.4).

Corollary 1.3. *For every $\delta < 25/228 = 0.1096\ldots$ and all large x, there are $\gg_\delta x^{1/2+\delta}$ integers $n \leq x$ such that n^3+1 is a sum of two squares, and $\gg_\delta x^{1/2+\delta}$ integers $n \leq x$ such that $n-1$, n and $n+1$ are all sums of two squares; the same holds for $C_a(x)$ and $S_{a,b}(x)$ with any fixed a, b. If Selberg's eigenvalue conjecture holds, every $\delta < 1/8$ is admissible.*

The proofs use the large sieve inequality of Pascadi for exceptional Maass forms [7, Theorem 2], through [4, Theorem 6.1(a)]. With only the inequalities of Deshouillers and Iwaniec [2] we obtain the following.

Theorem 1.4. *Theorems 1.1 and 1.2 hold with δ_θ^{**} in place of δ_θ^*, without using* [7, Theorem 2].

θ	[4, Thm. 1.1]	Theorems 1.1, 1.2	[4, Thm. 1.3]	Theorem 1.4
1/4 (Selberg)	$1/14 = 0.0714$	$1/12 = 0.0833$	$1/16 = 0.0625$	$1/14 = 0.0714$
7/64 (Kim–Sarnak)	$25/278 = 0.0899$	$25/228 = 0.1096$	$25/292 = 0.0856$	$25/242 = 0.1033$
0 (Selberg's conjecture)	1/10	1/8	1/10	1/8

TABLE 1. Exponents δ in lower bounds $\gg x^{1/2+\delta}$. The first and third columns are for three-term patterns only. Theorems 1.1–1.4 hold for every δ below the value shown.

1.1. **The idea.** We recall the families of [4]. Write the pattern as $\{P, P+B, T\}$ with B odd, $T = P+A$ and $Q = 2P+B$ [4, Lemma 2.3]. If $2dQ = u^2+d^2+B^2$, then $4P(P+B) = (Q-d)^2+u^2$, so for a prime $d \equiv 1 \pmod 4$ the numbers P and $P+B$ are sums of two squares, and the third number satisfies $4dT = u^2 + m_d$ with $m_d = (d+c)^2 + 4A(B-A)$, $c = 2A-B$. For fixed d, T runs through the values of a quadratic polynomial, and about $\sqrt{x/d}$ of them are at most x. After the hyperbola method, the r-weighted count of these values leads to Type I sums for the roots of a quadratic congruence modulo dk. The Type I estimate [4, Theorem 6.1] bounds them by

$$X^{1/2}H^{1/4}\Big(1 + \frac{X}{dH^{1/2}}\Big)^{\theta}, \tag{1.2}$$

up to factors x^ε, where $X \asymp \sqrt{dx}$ is the length of the polynomial sequence and $-4EH$ (with $E \in \{4,8\}$, $H \asymp m_d$) is the discriminant. The level $q = 4Ed$ enters only through the exceptional factor. The main term is $\asymp X/d$, and since $H \asymp d^2$ the comparison of (1.2) with X/d restricts d to $d \leq x^{(1-2\theta)/(5-6\theta)}$.

The loss $H^{1/4}$ comes from bounding the number of pairs of Heegner points at bounded distance without using the level structure [4, Remark 13.1]. It depends on the discriminant alone. So one should look for families whose discriminant is small compared with their level. They are obtained by inserting a square factor r^2. If

$$2rdQ = u^2 + d^2 + r^2B^2,$$

then $(rQ-d)^2+u^2 = r^2(Q^2-B^2) = 4r^2P(P+B)$, and since r^2 is a square, $P(P+B)$ is still a sum of two squares. The third number now satisfies $4rdT = u^2+M_{r,d}$ with $M_{r,d} = (d+cr)^2+4A(B-A)r^2$ (Lemma 2.2). The level is rd and the discriminant is $\asymp (d+r)^2$. For primes $r \asymp d \asymp t$ the level is $\asymp t^2$, the discriminant is $\asymp t^2$, each family contributes $\asymp \sqrt{x}/t$ values, and (1.2) is smaller than the main term as long as $t \le x^{\delta_\theta^*-\varepsilon}$. There are $\asymp t^2/(\log t)^2$ such families, so they produce $\asymp \sqrt{x}\,t$ values, with multiplicity $x^{o(1)}$ (Lemma 2.3). This gives the exponent δ_θ^*.

For n^3+a^3 the same mechanism applies to the factorisation (1.1). The quadratic factor is made a sum of two squares by Lemma 2.1, and m is detected. Here the case $r = 1$ gives the analogue of [4], with exponent $(1-2\theta)/(10-12\theta)$ (Remark 7.1).

In terms of Gaussian integers, $(rQ-d)+iu = \varpi^2 w$, where $\varpi \mid r$ is a Gaussian prime and w is a representation of $4P(P+B)$. The families of [4] collect the representations of $4P(P+B)$ close to the positive real axis; the rescaled families collect those close to the directions $\arg \varpi^{-2}$, which are the rational points of the unit circle with denominator r.

1.2. **What is used.** From [4] we use the hyperbola identity [4, Lemma 5.1], the 2-adic set-up for three-term patterns [4, Lemma 2.3, Proposition 3.2], and the Type I estimate [4, Theorem 6.1]. Theorem 3.1 below extends the latter to levels $q = 4Ed$ in which d is a product of at most two primes and to $E = 1$; the case $E = 1$ is used for n^3+a^3, where the variable ℓ runs over the odd integers. For $a \equiv 3 \pmod 4$ this is forced by the 2-adic conditions (Remark 5.1); for $a \equiv 1 \pmod 4$ we use odd ℓ as well, so that both cases are treated in the same way. The proof consists of checking the few places where [4] uses that d is prime or that $E \in \{4, 8\}$. Section 4 redoes the asymptotic evaluation of [4, §§5,7] for general families, Sections 5 and 6 prove the theorems, and Section 8 reports numerical checks.

Provenance. This note was written by Claude (Anthropic). The family construction of [4], that is, making two of the three numbers sums of two squares through the identity $2dQ = u^2+d^2+B^2$ and counting the third along the resulting quadratic polynomial, is due to Siddharth Iyer; that is his only contribution to the present note. The rescaled families, their analogue for n^3+a^3, the extension of the Type I estimate, the arguments and the computations below were developed by Claude. The note has not been refereed.

Notation. χ is the non-principal character modulo 4, so $r(n) = 4\sum_{k\mid n}\chi(k)$ for $n \ge 1$. We write v_2 for the 2-adic valuation, $\omega(n)$ for the number of distinct prime factors, and $\left(\frac{\cdot}{\cdot}\right)$ for the Legendre or Jacobi symbol. The smooth functions F (supported in $[\frac12, 1]$, values in $[0,1]$, not identically zero), W (with $W(t)+W(1/t) = 1$, $W = 1$ on $(0, \frac12]$, $W = 0$ on $[2,\infty)$) and ψ_0 are those of [4, §§4,5,7], and $c_F = \int_0^1 F(t)t^{-1/2}\,dt$. The letter r denotes a prime as well as the function $r(n)$, which is always written with its argument. The letter L denotes the level of a family; it has nothing to do with the quantity L in [4, §§9–12]. Throughout, θ is a fixed admissible exponent, $\varepsilon, \eta > 0$ are small, and implied constants may depend on a, b, ε, η and the fixed smooth functions.

2. Rescaled families

Lemma 2.1 (Cubes)**.** *Let a, r, D, ℓ, m be integers such that $w = D + 3ar$ is even, and put $H = (w/2)^2 - 3a^2r^2$. If*

$$rD\,m = \ell^2 + H,$$

then

$$(2\ell)^2 + (2rm - 3ar - D)^2 = 4r^2 G_a(m). \tag{2.1}$$

In particular, if $r \ne 0$, then $G_a(m) \in \mathcal{S}$; and if moreover $m \ge 1$ and $m \in \mathcal{S}$, then $n = m - a$ satisfies $n^3 + a^3 \in \mathcal{S}$.

Proof. We have $(2rm-3ar-D)^2=(2rm-w)^2=4r^2m^2-4rmw+w^2$ and $4\ell^2=4rDm-4H=4rDm-w^2+12a^2r^2$. Adding, and using $w-D=3ar$, gives $4r^2m^2-12ar^2m+12a^2r^2=4r^2G_a(m)$. Since $\mathcal{S}$ is closed under multiplication and $p^{2k}\in\mathcal{S}$ for every prime p, an integer $z\ge1$ lies in $\mathcal{S}$ if and only if $4r^2z$ does; and $G_a(m)>0$ because its discriminant $-3a^2$ is negative. The last assertion follows from (1.1). □

Lemma 2.2 (Three-term patterns)**.** *Let A,B be integers with B odd, put $c=2A-B$, and let r,d,u,T be integers such that*

$$4rdT=u^2+M_{r,d},\qquad M_{r,d}=(d+cr)^2+4A(B-A)r^2.$$

Put $P=T-A$ and $Q=2T-c=2P+B$. Then

$$2rdQ=u^2+d^2+r^2B^2\qquad\text{and}\qquad 4r^2P(P+B)=(rQ-d)^2+u^2.$$

If $r\neq0$, d is a prime with $d\equiv1\pmod4$, and $P\ge1$, $P+B\ge1$, then $P\in\mathcal{S}$ and $P+B\in\mathcal{S}$.

Proof. Since $c^2+4A(B-A)=(2A-B)^2+4AB-4A^2=B^2$, we get $2rdQ=4rdT-2rcd=u^2+(d+cr)^2-2rcd+4A(B-A)r^2=u^2+d^2+r^2B^2$. Then $(rQ-d)^2+u^2=r^2Q^2-2rdQ+d^2+u^2=r^2(Q^2-B^2)$, and $Q^2-B^2=4P(P+B)$. So $4r^2P(P+B)\in\mathcal{S}$, hence $P(P+B)\in\mathcal{S}$. Let $\ell'\equiv3\pmod4$ be a prime. If ℓ' divides at most one of P and $P+B$, then $v_{\ell'}(P)$ and $v_{\ell'}(P+B)$ are even. If ℓ' divides both, then $\ell'\mid B$ and $\ell'\mid Q$, so $\ell'\mid u^2+d^2$ by the first identity, hence $\ell'\mid d$, which is impossible since $d\equiv1\pmod4$ is prime. □

For $r=1$ this is [4, Lemmas 2.4 and 2.5]. The next lemma controls how often one integer arises from different families.

Lemma 2.3 (Multiplicity)**.** *Let $N_0\ge1$ and $R\ge2$ be integers and $0<\varphi_0\le1/(16R^2)$. Let $\mathcal{Z}$ be the set of pairs (r,z) such that $r\le R$ is a prime with $r\equiv1\pmod4$, $z\in\mathbb{Z}[i]$, $|z|^2=r^2N_0$, $z\notin r\mathbb{Z}[i]$, and $|\arg z|\le\varphi_0$. Then $\#\mathcal{Z}\le r(N_0)$.*

Proof. Let $(r,z)\in\mathcal{Z}$ and write $r=\pi\bar\pi$ with a Gaussian prime π; then π and $\bar\pi$ are not associate. Since $v_\pi(z)+v_{\bar\pi}(z)=v_\pi(z)+v_\pi(\bar z)=v_\pi(r^2N_0)\ge2$ and $z\notin r\mathbb{Z}[i]$, one of $v_\pi(z)$, $v_{\bar\pi}(z)$ is 0 and the other is at least 2. Hence $z=\varpi^2w$ with $\varpi\in\{\pi,\bar\pi\}$ and $|w|^2=N_0$. Choose such a factorisation for every element of $\mathcal{Z}$. We claim that two elements (r_1,z_1), (r_2,z_2) with the same w coincide; this proves the lemma, since there are $r(N_0)$ possible w.

So let $z_i=\varpi_i^2w$ with $|\varpi_i|^2=r_i$. Then $\alpha=\varpi_1^2\bar\varpi_2^2\in\mathbb{Z}[i]$ has $|\alpha|=r_1r_2\le R^2$ and $\arg\alpha\equiv\arg z_1-\arg z_2\pmod{2\pi}$. Hence $|\operatorname{Im}\alpha|\le R^2\cdot2\varphi_0<1$, so α is real, and $\alpha>0$ since $|\arg z_1-\arg z_2|<\pi/2$. Therefore $\varpi_1^2\bar\varpi_2^2=r_1r_2=\varpi_1\bar\varpi_1\varpi_2\bar\varpi_2$, that is, $\varpi_1\bar\varpi_2=\bar\varpi_1\varpi_2$. If $r_1\neq r_2$, then ϖ_1 divides $\bar\varpi_1\varpi_2$ but neither factor, a contradiction. So $r_1=r_2$ and ϖ_2 is associate to ϖ_1 or to $\bar\varpi_1$. In the second case $\varpi_2=\epsilon\bar\varpi_1$ with a unit ϵ, and the relation gives $\varpi_1^2=(\epsilon/\bar\epsilon)\bar\varpi_1^2$, which is impossible. In the first case $\varpi_2=\epsilon\varpi_1$ and the relation gives $\epsilon=\bar\epsilon$, so $\epsilon=\pm1$, $\varpi_2^2=\varpi_1^2$ and $z_2=z_1$. □

3. The Type I estimate at composite level

Consider the following hypothesis, which differs from [4, (H)] only in the first and third items.

(H*) $E\in\{1,4,8\}$; $H\ge1$ is odd; $d\ge3$ is odd and squarefree with $\omega(d)\le2$ and $(d,H)=1$; $N=EH$ is neither a square nor three times a square; $q=4Ed$; $\kappa\in\{1,3\}$; $\Delta\ge1$, $X\ge N^{1/2}$ and $1\le K\le qX$; $\psi_1,\psi_2:\mathbb{R}\to\mathbb{C}$ are smooth, ψ_1 is supported in $[1,2]$ and ψ_2 in $[-1,1]$, and $\|\psi_i^{(\nu)}\|_\infty\le C_\nu\Delta^\nu$ for all $\nu\ge0$.

Under (H*) let $\varrho_{E,H}(\mu) = \#\{\nu \bmod \mu :\ E\nu^2 + H \equiv 0 \pmod{\mu}\}$ and

$$\Xi = \sum_{\mu \equiv \kappa d \pmod{4d}} \psi_1\Big(\frac{\mu}{K}\Big)\Big(\sum_{\substack{\ell \in \mathbb{Z} \\ \mu \mid E\ell^2 + H}} \psi_2\Big(\frac{\ell}{X}\Big) - \frac{\varrho_{E,H}(\mu)}{\mu}\, X \int_{\mathbb{R}} \psi_2(t)\, dt\Big), \tag{3.1}$$

as in [4, (6.1)].

Theorem 3.1. *Assume* (H*). *For every $\eta > 0$ there is $C(\eta)$ such that*

$$\Xi \ll \Delta^{C(\eta)} (qXH)^{\eta}\, X^{1/2} H^{1/4} \Big(1 + \frac{X}{dH^{1/2}}\Big)^{\theta},$$
$$\Xi \ll \Delta^{C(\eta)} (qXH)^{\eta}\, X^{1/2} H^{1/4} \Big(1 + \frac{K}{dH^{1/2}}\Big)^{\theta},$$

where the implied constants depend only on η and the C_ν. The second bound is proved without [7, Theorem 2].

Proof. For $E \in \{4, 8\}$ and d prime this is [4, Theorem 6.1]. Its proof, in [4, §§8–12], uses the primality of d and the condition $E \in \{4, 8\}$ only at the following places; everywhere else it uses only that $q = 4Ed$, that d is odd with $(d, H) = 1$, and that N is neither a square nor three times a square. In particular, the large sieve inequalities [4, Lemmas 10.2–10.5], the spectral expansion and the pre-trace formula hold for $\Gamma_0(q)$ with any q, and in [4, §§9,11] the integer d enters only through the level q and the congruence $\mu \equiv \kappa d \pmod{4d}$, not through its primality.

(i) *The sets $\mathcal{Q}^\kappa$* [4, Lemma 8.2]. The set $\mathcal{Q}^\kappa = \{\mathfrak{g} \in \mathcal{Q}_N :\ \mathfrak{b} \equiv 0\ (E),\ \mathfrak{c} \equiv \kappa E d\ (4Ed)\}$ is defined in the same way; for $E = 1$ the first condition is empty. The bijection of part (a) is unchanged. The proof of the invariance in part (b) uses only that $q \mid \gamma_3$, that γ_4 is odd, and that E divides γ_3, $\mathfrak{b}$ and $\mathfrak{c}$ and Ed divides $\mathfrak{c}$; this holds verbatim.

(ii) *Cosets* [4, Lemma 8.3]. By the Chinese remainder theorem $\mathbb{P}^1(\mathbb{Z}/q\mathbb{Z}) = \mathbb{P}^1(\mathbb{Z}/4E\mathbb{Z}) \times \prod_{p \mid d} \mathbb{P}^1(\mathbb{F}_p)$. For each prime $p \mid d$ the form $\varphi_{\mathfrak{g}}$ does not vanish identically modulo p, since otherwise $p^2 \mid N$, which is impossible because p is odd and $p \nmid H$; so it has at most two zeros in $\mathbb{P}^1(\mathbb{F}_p)$. Since $\#\mathbb{P}^1(\mathbb{Z}/4E\mathbb{Z}) = 6E$, the bound [4, (8.2)] becomes $\#\{\tau :\ \tau \diamond \mathfrak{g} \in \mathcal{Q}\} \le 2^{\omega(d)} \cdot 6E \le 24E \le 192$.

(iii) *Pairs* [4, Lemma 8.4]. With (ii), the constants 96 and 192 there become 192 and 384. The bounds $I \ll N^{1/2+\eta}$ and $\mathcal{P} \ll N^{1+\eta}$ are unchanged.

(iv) *Sizes* [4, (12.1)]. There the inequalities $\frac{1}{16qX} \le MY \le \frac{(qX)^\eta}{2}$ are derived from $4 \le E \le 8$. For $1 \le E \le 8$ the lower bound is unchanged, since $Y = N^{1/2}/(EK) \ge 1/(8qX)$, while the upper bound becomes $MY \le 2(qX)^\eta/E \le 2(qX)^\eta$. We therefore define $V = 4(qX)^\eta/(MY)$ instead of $(qX)^\eta/(MY)$, so that again $V \ge 1$ (indeed $V \ge 2$). Lemma 11.1 of [4] is then applied with this V; the error term $(2/y_0)e^{-y_0 V/2}$ with $y_0 = \pi MY$ only becomes smaller. The factors $\mathfrak{E}_{\mathrm{P}}$ and $\mathfrak{E}_{\mathrm{DI}}$ change by at most $4^\theta \le 2$, and the bound $(MY)^{3/2} \le (qX)^{3\eta/2}$ used in [4, §12.4] becomes $(MY)^{3/2} \le 3(qX)^{3\eta/2}$.

(v) *Lemma 12.1 of* [4] *and the completion of the proof.* With the new V the quantities α_1 and β_1 in the proof of [4, Lemma 12.1] are multiplied by 4, which changes the constants only. The inequality $EX/(q\sqrt{N}) = X/(4d\sqrt{N}) \le X/(dH^{1/2})$ used there holds because $N = EH \ge H$; and the final step uses $N \le 8H$, which holds for $E \le 8$. □

4. Weighted counts along a family

We now evaluate the r-weighted count along a general family, following [4, §§5,7].

Setting (F).

(F1) $E \in \{1, 4, 8\}$, $H \ge 1$ is odd, and $N = EH$ is neither a square nor three times a square.

(F2) L is an odd prime or a product of two distinct odd primes, $(L,H)=1$, and $-EH$ is a quadratic residue modulo every prime factor of L.
(F3) Either $\Lambda=\mathbb{Z}$, and then $c_\Lambda=1$; or $E=1$ and $\Lambda=2\mathbb{Z}+1$, and then $c_\Lambda=\frac12$.
(F4) $s\ge s_0\ge 0$ are integers such that for every $\ell\in\Lambda$ with $L\mid E\ell^2+H$ the number $2^{-s_0}(E\ell^2+H)$ is an odd integer and $(E\ell^2+H)/(2^{s_0}L)\equiv 1 \pmod 4$.

Put $\kappa_0=2^{s-s_0}$ and $T(y)=\kappa_0(Ey^2+H)/L$ for $y\in\mathbb{R}$. By (F4), for $\ell\in\Lambda$ with $L\mid E\ell^2+H$ the number $T(\ell)$ is an integer of the form $2^sT'$ with $T'=(E\ell^2+H)/(2^{s_0}L)\equiv 1 \pmod 4$. We put

$$\Sigma(x)=\sum_{\substack{\ell\in\Lambda\\ L\mid E\ell^2+H}} r\big(T(\ell)\big)\,F\Big(\frac{T(\ell)}{x}\Big).$$

Lemma 4.1 (Roots)**.** *Assume* (F1)–(F2) *and write* $\varrho=\varrho_{E,H}$.

(a) *For* $p\mid L$ *and* $i\ge 1$, $\varrho(p^i)=2$.
(b) *For an odd prime* $p\nmid L$ *and* $i\ge 1$, $\varrho(p^i)=1+\left(\frac{-EH}{p}\right)$ *if* $p\nmid H$, *and in general* $\varrho(p^i)\le 2p^{\min(i,v_p(H))/2}$.
(c) *For odd* $k\ge 1$, $\varrho(Lk)=2^{\omega(L)}f(k)$, *where* f *is the multiplicative function with* $f(p^i)=1$ *for* $p\mid L$, $f(p^i)=\varrho(p^i)$ *for odd* $p\nmid L$, *and* $f(2^i)=0$ *for* $i\ge 1$.

Proof. For odd p the integer E is invertible modulo p^i, and $E\nu^2+H\equiv 0$ is equivalent to $(E\nu)^2\equiv -EH$. Part (a) follows from Hensel's lemma, since $p\nmid 2EH$ and $-EH$ is a non-zero square modulo p. Part (b) is [4, Lemma 5.2(b)] with m replaced by EH. Part (c) follows from (a) and the Chinese remainder theorem. □

For odd $n\ge 1$ let $\psi(n)=\left(\frac{EH}{n}\right)$ (Jacobi symbol), and $\psi(n)=0$ for even n. Put $A_f(y)=\sum_{k\le y}\chi(k)f(k)$ and $\mathfrak{L}=\sum_{k\ge1}\chi(k)f(k)/k$.

Lemma 4.2 (Singular series)**.** *Assume* (F1)–(F2).

(a) ψ *is a non-principal real Dirichlet character modulo* $4EH$.
(b) $A_f(y)\ll_\varepsilon y^{1/2+\varepsilon}(EH)^{1/2+\varepsilon}L^\varepsilon$ *for* $y\ge 1$.
(c) *The series* $\mathfrak{L}$ *converges,* $|\mathfrak{L}|\ll_\varepsilon (EH)^{1/2+\varepsilon}L^\varepsilon$, *and* $\mathfrak{L}=L(1,\chi)L(1,\psi)\mathcal{P}$ *with*

$$\mathcal{P}=\prod_{p\nmid 2LH}\Big(1-\frac{1}{p^2}\Big)\prod_{p\mid L}\Big(1-\frac{\chi(p)}{p}\Big)\prod_{p\mid H}\Big(1-\frac{\chi(p)}{p}\Big)\sum_{i\ge0}\frac{\chi(p)^i\varrho(p^i)}{p^i}\ \ge\ \frac{6}{\pi^2}\,\frac{\phi(LH)}{LH}.$$

(d) $\mathfrak{L}\gg_\varepsilon (LH)^{-\varepsilon}$. *The implied constant is ineffective.*

Proof. This is [4, Lemma 5.3] with m replaced by EH and d replaced by L; we indicate the differences. Part (a) is proved as [4, Lemma 5.3(a)]: $EH=2^eH$ with $e\in\{0,2,3\}$, and EH is not a square by (F1). For (b) and (c) we compare the Euler factors of χf and $\chi*\psi$, as power series in a variable Y. At $p=2$ both are 1. For odd $p\nmid LH$ they are as in [4], because $\psi(p)=\left(\frac{EH}{p}\right)=\chi(p)\left(\frac{-EH}{p}\right)$. For $p\mid L$ we have $\psi(p)=\chi(p)\left(\frac{-EH}{p}\right)=\chi(p)$ by (F2), so the factor of $\chi*\psi$ is $(1-\chi(p)Y)^{-2}$, while that of χf is $(1-\chi(p)Y)^{-1}$ by Lemma 4.1(c); the quotient is $1-\chi(p)Y$. (In [4], $d\equiv 1 \pmod 4$ and the quotient is $1-Y$.) For odd $p\mid H$ the factors are as in [4]. Hence $\chi f=(\chi*\psi)*\beta$ with β supported on odd integers, $\beta(p^2)=-1$ for $p\nmid 2LH$, $\beta(p)=-\chi(p)$ for $p\mid L$, and $\beta(p^i)=\chi(p)^i(\varrho(p^i)-\varrho(p^{i-1}))$ for $p\mid H$, all other values on prime powers being 0. As in [4], $\sum_n|\beta(n)|n^{-1/2-\varepsilon}\ll_\varepsilon (LH)^\varepsilon$, and the Pólya–Vinogradov inequality gives $\sum_{n\le y}(\chi*\psi)(n)\ll y^{1/2}(EH)^{1/2+\varepsilon}$; this gives (b), and (c) follows as in [4], including the lower bound for the factors at $p\mid H$. Part (d) follows from (c), $L(1,\chi)=\pi/4$, $\phi(n)/n\gg(\log\log 3n)^{-1}$ and Siegel's theorem, as in [4, Lemma 5.3(d)]. □

Proposition 4.3. *Assume* (F), *let* $0<\varepsilon<1/100$, *and suppose that* $LH\le x$, $H\le x^{3/4}L$ *and* $x\ge x_0(\varepsilon,s)$. *Then*

$$\Sigma(x)=2^{\omega(L)+3}\,c_\Lambda\,c_F\,\mathfrak{L}\sqrt{\frac{x}{\kappa_0 EL}} + O\Big(x^{1/4+\varepsilon}\Big(\frac{H}{L}\Big)^{1/2}\Big)+O\big(x^{1/4+\theta/2+\varepsilon}(LH)^{1/4-\theta/2}\big).$$

If the second bound of Theorem 3.1 is used in place of the first, the same holds with $x^{1/4+\theta/2+\varepsilon}L^{1/4}H^{1/4-\theta/2}$ *in place of the last error term. The implied constants depend on* ε, s *and the fixed smooth functions.*

4.1. **Set-up.** Let $X>0$ be defined by $T(X)=x$. Since $\kappa_0 H\le \kappa_0 x^{3/4}L\le Lx/4$ for large x,

$$\frac{Lx}{2\kappa_0 E}\le X^2\le \frac{Lx}{\kappa_0 E}. \tag{4.1}$$

In particular $X^2\ge 16EH$, and hence $X\ge N^{1/2}$ and $X/2\ge (4H)^{1/2}$, for large x, because $H\le x^{3/4}L$. For $y>0$ and $t\in\mathbb{R}$ put $\Phi_y(t)=F(T(t)/x)\,W\big(2^{s/2}y/\sqrt{T(t)}\big)$.

Let $\ell\in\Lambda$ with $L\mid E\ell^2+H$. Then $T(\ell)=2^sT'$ with $T'\equiv 1 \pmod 4$, so [4, Lemma 5.1] gives $r(T(\ell))=8\sum_{k\mid T(\ell)}\chi(k)W(2^{s/2}k/\sqrt{T(\ell)})$. For odd k we have $k\mid T(\ell)$ if and only if $k\mid T'$, that is, if and only if $Lk\mid E\ell^2+H$, since $E\ell^2+H=2^{s_0}LT'$. Hence

$$\Sigma(x)=8\sum_{k\ \text{odd}}\chi(k)\sum_{\substack{\ell\in\Lambda\\ Lk\mid E\ell^2+H}}\Phi_k(\ell), \tag{4.2}$$

and only $k<k_+:=2^{1-s/2}\sqrt{x}$ contribute. For odd μ, the number of $\ell\in\Lambda$ in a residue class modulo μ has density c_Λ/μ. Accordingly we put

$$\mathcal{M}=8\sum_{k\ \text{odd}}\chi(k)\,c_\Lambda\frac{\varrho(Lk)}{Lk}\int_{\mathbb{R}}\Phi_k(t)\,dt,\qquad \mathcal{E}=\Sigma(x)-\mathcal{M}.$$

4.2. **The main term.**

Lemma 4.4. $\mathcal{M}=2^{\omega(L)+3}\,c_\Lambda\,c_F\,\mathfrak{L}\sqrt{x/(\kappa_0 EL)}+O\big(x^{1/4+\varepsilon}(H/L)^{1/2}\big)$.

Proof. By Lemma 4.1(c), and since f vanishes on even integers,

$$\mathcal{M}=\frac{2^{\omega(L)+3}c_\Lambda}{L}\int_{\mathbb{R}}F\Big(\frac{T(t)}{x}\Big)G\big(2^{-s/2}\sqrt{T(t)}\big)\,dt,\qquad G(y)=\sum_{k\ge1}\frac{\chi(k)f(k)}{k}W\Big(\frac{k}{y}\Big).$$

As in [4, Lemma 7.2], partial summation and Lemma 4.2(b) give $\mathfrak{L}-G(y)\ll y^{-1/2+\varepsilon}(EH)^{1/2+\varepsilon}L^{\varepsilon}$ for $y\ge1$. For $t>0$ we substitute $T=T(t)$. Then $dT/dt=2\kappa_0 Et/L$ and $\kappa_0 Et=\sqrt{\kappa_0 E}\sqrt{LT-\kappa_0 H}$, so, the integrand being even,

$$\mathcal{M}=\frac{2^{\omega(L)+3}c_\Lambda}{\sqrt{\kappa_0 E}}\int_{x/2}^{x}\frac{F(T/x)\,G(2^{-s/2}\sqrt{T})}{\sqrt{LT-\kappa_0 H}}\,dT.$$

On this range $LT-\kappa_0 H\ge LT/2$, since $\kappa_0 H\le Lx/4$. Replacing G by $\mathfrak{L}$ costs

$$\ll\int_{x/2}^{x}x^{-1/4+\varepsilon}(EH)^{1/2+\varepsilon}L^{\varepsilon}(LT)^{-1/2}\,dT\ll x^{1/4+2\varepsilon}(H/L)^{1/2},$$

using $EH\le 8LH\le 8x$. Replacing $(LT-\kappa_0 H)^{-1/2}$ by $(LT)^{-1/2}$ costs $\ll|\mathfrak{L}|\,xH(Lx)^{-3/2}$, which is $\ll H^{3/2+\varepsilon}L^{-3/2+\varepsilon}x^{-1/2}\le x^{1/4+2\varepsilon}(H/L)^{1/2}$, because $H/L\le x^{3/4}$. Finally $\int F(T/x)(LT)^{-1/2}\,dT=c_F\sqrt{x/L}$. Renaming ε gives the lemma. ∎

4.3. **The error term.**

Lemma 4.5. *We have $\mathcal{E} \ll x^{1/4+\theta/2+\varepsilon}(LH)^{1/4-\theta/2}$. With the second bound of Theorem 3.1 in place of the first, $\mathcal{E} \ll x^{1/4+\theta/2+\varepsilon}L^{1/4}H^{1/4-\theta/2}$.*

Proof. We follow [4, Lemma 7.3]. With $\mathcal{K}$ and ψ_0 as there,

$$\mathcal{E} = 8\sum_{K\in\mathcal{K}}\sum_{\kappa\in\{1,3\}}\chi(\kappa)\,\mathcal{E}(K,\kappa),$$

$$\mathcal{E}(K,\kappa) = \sum_{k\equiv\kappa\ (4)}\psi_0\Big(\frac{k}{K}\Big)\Big(\sum_{\substack{\ell\in\Lambda\\ Lk\mid E\ell^2+H}}\Phi_k(\ell) - c_\Lambda\frac{\varrho(Lk)}{Lk}\int_{\mathbb{R}}\Phi_k(t)\,dt\Big),$$

and only $O(\log x)$ values of K contribute. Write $\mu = Lk$, so that $k \equiv \kappa \pmod 4$ if and only if $\mu \equiv \kappa L \pmod{4L}$. Let $\psi_2(\sigma) = F(T(X\sigma)/x)$; since $T(X\sigma)/x = \alpha\sigma^2 + 1 - \alpha$ with $\alpha = EX^2/(EX^2+H) \in [\frac12, 1]$, the function ψ_2 is supported in $[-1,1]$ with bounded derivatives. Exactly as in [4, Lemma 7.3], with $k_- = 2^{-3-s/2}\sqrt{x}$, one has $\mathcal{E}(K,\kappa) = \Xi_\Lambda$ for $K \le k_-$ and $\mathcal{E}(K,\kappa) = \frac{1}{2\pi i}\int_{(1)}\widetilde{W}(w)\,\Xi_{\Lambda,w}\,dw$ for $k_- < K < k_+$, where Ξ_Λ (respectively $\Xi_{\Lambda,w}$) is the analogue of (3.1) with the sum over $\ell \in \Lambda$, the factor c_Λ in the main term, d replaced by L, K replaced by LK, and the weights ψ_1, ψ_2 (respectively $\psi_{1,w}, \psi_{2,w}$) of [4, Lemma 7.3], with $\Delta = 1$ (respectively $\Delta = 1 + |w|$).

If $\Lambda = \mathbb{Z}$, then Ξ_Λ is the sum (3.1). If $\Lambda = 2\mathbb{Z}+1$, then $E = 1$, and since μ is odd,

$$\sum_{\substack{\ell\ \text{odd}\\ \mu\mid\ell^2+H}}\psi_2\Big(\frac{\ell}{X}\Big) = \sum_{\substack{\ell\in\mathbb{Z}\\ \mu\mid\ell^2+H}}\psi_2\Big(\frac{\ell}{X}\Big) - \sum_{\substack{\ell'\in\mathbb{Z}\\ \mu\mid 4\ell'^2+H}}\psi_2\Big(\frac{\ell'}{X/2}\Big),$$

while $\varrho_{4,H}(\mu) = \varrho_{1,H}(\mu)$ and $\frac12 X = X - \frac{X}{2}$. Hence Ξ_Λ is the difference of the sum (3.1) for $(E,H,X) = (1,H,X)$ and for $(4,H,X/2)$.

We apply Theorem 3.1 with $d = L$. The hypotheses (H*) hold: (F1) and (F2) give the conditions on E, H, d and N (note that $4H$ is neither a square nor three times a square when H is not); $X \ge N^{1/2}$ and $X/2 \ge (4H)^{1/2}$ hold for large x, as noted after (4.1); and $LK \le Lk_+ = 2^{1-s/2}L\sqrt{x} \le qX$ by (4.1), since $E, L \ge 1$, and similarly with $X/2$ and $q = 16L$. By (4.1) we have $X^{1/2} \ll (Lx)^{1/4}$ and $X/(LH^{1/2}) \le (x/(LH))^{1/2}$, and $(x/(LH))^{1/2} \ge 1$ since $LH \le x$; so $1 + X/(LH^{1/2}) \le 2(x/(LH))^{1/2}$. The first bound of Theorem 3.1 therefore gives

$$\Xi_\Lambda,\ \Xi_{\Lambda,w} \ll (1+|w|)^C x^\varepsilon (Lx)^{1/4}H^{1/4}\Big(\frac{x}{LH}\Big)^{\theta/2} = (1+|w|)^C x^{1/4+\theta/2+\varepsilon}(LH)^{\frac14-\frac{\theta}{2}}.$$

With the second bound we use $LK/(LH^{1/2}) \le k_+/H^{1/2} \ll (x/H)^{1/2}$ and $H \le x$ instead, and obtain

$$\Xi_\Lambda,\ \Xi_{\Lambda,w} \ll (1+|w|)^C x^{1/4+\theta/2+\varepsilon}L^{1/4}H^{1/4-\theta/2}.$$

Integrating against $|\widetilde{W}(w)|$ and summing over K and κ proves the lemma. □

Proposition 4.3 follows from Lemmas 4.4 and 4.5.

5. Sums of two squares along $n^3 + a^3$

Throughout this section a is odd. Let D_a be the residue class modulo 8 given by

$$D_a \equiv 4 - 3a \pmod 8 \quad \text{if } a \equiv 1 \pmod 4, \qquad D_a \equiv -3a \pmod 8 \quad \text{if } a \equiv 3 \pmod 4.$$

Note that $D_a \equiv a \pmod 4$. For integers r, D with $D + 3ar$ even put

$$H_{r,D} = \Big(\frac{D+3ar}{2}\Big)^2 - 3a^2r^2.$$

Remark 5.1. In [4] the variable runs over a lattice $2^j\mathbb{Z}$. For $a \equiv 3 \pmod 4$ this is impossible here. Indeed, $G_a(m)$ is odd for every m, so in (2.1) we have $v_2(2rm - 3ar - D) = 1$, and if $4 \mid \ell$ then $4r^2 G_a(m)/(2rm - 3ar - D)^2 \equiv 1 \pmod{16}$; hence $G_a(m) \equiv 1 \pmod 8$. If m is odd, then $G_a(m) \equiv am \pmod 4$, so $m \equiv a \equiv 3 \pmod 4$. If m is even, then $4 \nmid m$ (otherwise $G_a(m) \equiv 3 \pmod 4$), and with $m = 2m''$ we get $G_a(m) \equiv 7 - 6am'' \pmod 8$, so $m'' \equiv 3 \pmod 4$. In both cases $r(m) = 0$. This is why ℓ runs over the odd integers below, which requires $E = 1$ in Theorem 3.1.

Lemma 5.2 (2-adic set-up)**.** *Let $r \equiv 1 \pmod 8$ and $D \equiv D_a \pmod 8$, and put $H = H_{r,D}$. Then $4 \mid D + 3ar$ and $H \equiv 1 \pmod 4$. Moreover, for every odd ℓ with $rD \mid \ell^2 + H$, we have $v_2(\ell^2 + H) = 1$ and $(\ell^2 + H)/(2rD) \equiv 1 \pmod 4$.*

Proof. Let $w = D + 3ar$. Since $r \equiv 1 \pmod 8$, $w \equiv D + 3a \pmod 8$, so $w \equiv 4 \pmod 8$ if $a \equiv 1 \pmod 4$ and $w \equiv 0 \pmod 8$ if $a \equiv 3 \pmod 4$. Write $w = 4y$; then y is odd if and only if $a \equiv 1 \pmod 4$. Since ar is odd, $a^2r^2 \equiv 1 \pmod 8$, so $H = 4y^2 - 3a^2r^2 \equiv 4y^2 - 3 \pmod 8$, which gives $H \equiv 1 \pmod 4$. For odd ℓ we have $\ell^2 \equiv 1 \pmod 8$, hence $\ell^2 + H \equiv 4y^2 - 2 \pmod 8$. So $v_2(\ell^2 + H) = 1$ and $(\ell^2 + H)/2 \equiv 2y^2 - 1 \pmod 4$, which is 1 if $a \equiv 1 \pmod 4$ and 3 if $a \equiv 3 \pmod 4$. Since $rD \equiv D \equiv a \pmod 4$, in both cases $(\ell^2 + H)/(2rD) \equiv 1 \pmod 4$. □

Definition 5.3. A pair (r, D) is *good* (for a) if r is a prime with $r \equiv 1 \pmod 8$; D is a prime with $D \equiv D_a \pmod 8$ and $\left(\frac{3}{D}\right) = 1$; $ar \le D \le 2ar$, $D \ne r$ and $D > 3a$; and $H_{r,D}$ is neither a square nor three times a square.

For a good pair put $L = rD$, $H = H_{r,D}$, $m(\ell) = (\ell^2 + H)/(rD)$ and

$$\Sigma_{r,D}(x) = \sum_{\substack{\ell \text{ odd} \\ rD \mid \ell^2 + H}} r\big(m(\ell)\big)\, F\Big(\frac{m(\ell)}{x}\Big).$$

Lemma 5.4. *For a good pair (r, D), the setting* (F) *holds with $E = 1$, $H = H_{r,D}$, $L = rD$, $\Lambda = 2\mathbb{Z} + 1$ and $s = s_0 = 1$, so that $\kappa_0 = 1$ and $T(\ell) = m(\ell)$. Moreover $a^2r^2 \le H \le \frac{13}{4}a^2r^2$ and $ar^2 \le L \le 2ar^2$.*

Proof. Lemma 5.2 gives that H is odd and gives (F4) with $s = s_0 = 1$; (F1) holds by definition, and (F3) holds. For (F2): $4H = (D + 3ar)^2 - 12a^2r^2 \equiv D^2 \pmod r$ and $r \nmid D$, so $r \nmid H$, and $-4H \equiv -D^2$ is a quadratic residue modulo r because $r \equiv 1 \pmod 4$. Modulo D we have $4H \equiv 9a^2r^2 - 12a^2r^2 = -3a^2r^2$, and $D \nmid 3ar$ because $D > 3a$ is a prime different from r; so $D \nmid H$, and $-4H \equiv 3a^2r^2$ is a quadratic residue modulo D because $\left(\frac{3}{D}\right) = 1$. Finally $\frac{D+3ar}{2} \in [2ar, \frac{5}{2}ar]$ gives the bounds for H. □

Proposition 5.5. *Let x be large and let $\mathcal{D}$ be a set of good pairs with $r \le x^{1/5}$ for all $(r, D) \in \mathcal{D}$. Then*

$$\sum_{(r,D)\in\mathcal{D}} \Sigma_{r,D}(x) \le C_a(x) \cdot \max_{x/2 \le m \le x} r(m)\, r\big(4G_a(m)\big) \ll_\varepsilon C_a(x)\, x^\varepsilon.$$

Proof. Let $(r, D) \in \mathcal{D}$ and let ℓ be odd with $rD \mid \ell^2 + H$ and with a non-zero term in $\Sigma_{r,D}(x)$. Then $m = m(\ell)$ satisfies $x/2 \le m \le x$ and $m \in \mathcal{S}$, and Lemma 2.1 (with $w = D + 3ar$, which is even) shows that $n = m - a$ satisfies $1 \le n \le x$ and $n^3 + a^3 \in \mathcal{S}$.

Fix such an n, hence m, and put $N_0 = 4G_a(m)$. To each (r, D, ℓ) as above with this m we attach $z = (2rm - 3ar - D) + 2i\ell \in \mathbb{Z}[i]$. By (2.1), $|z|^2 = r^2 N_0$, and $z \notin r\mathbb{Z}[i]$ because $\operatorname{Re} z \equiv -D \not\equiv 0 \pmod r$. From $\ell^2 < rDm \le 2ar^2 m$ and $\operatorname{Re} z \ge 2rm - 5ar \ge rm$ we get $|\arg z| \le 2\sqrt{2a/m} \le 4\sqrt{a/x} =: \varphi_0$. With $R = \lfloor x^{1/5} \rfloor$ we have $\varphi_0 \le 1/(16R^2)$ for large x, so Lemma 2.3 shows that there are at most $r(N_0)$ pairs (r, z). Since (r, z) determines $D = 2rm - 3ar - \operatorname{Re} z$ and $\ell = \operatorname{Im} z/2$, the integer n arises from at most $r(4G_a(m))$ triples (r, D, ℓ). As $0 \le F \le 1$, this proves the first inequality, and the second follows from $r(n) \ll_\varepsilon n^\varepsilon$. □

Proof of Theorems 1.1 and 1.4 for $C_a(x)$. First let a be odd. Let $\delta < \delta_\theta^*$, choose ϖ with $\delta < \varpi < \delta_\theta^*$, and then $\varepsilon > 0$ so small that $\varpi(2-2\theta) + 3\varepsilon < \frac14 - \frac{\theta}{2}$ and $\varpi - 3\varepsilon > \delta$. Put $R = x^{\varpi}$ and let $\mathcal{D}$ be the set of good pairs with $R < r \le 2R$.

For $(r, D) \in \mathcal{D}$ we have $LH \le 7a^3(2R)^4 \le x^{1/2}$, because $\varpi < 1/8$, and $H \le 4aL$. By Lemma 5.4, Proposition 4.3 applies with $\omega(L) = 2$, $c_\Lambda = \frac12$ and $\kappa_0 E = 1$, and by Lemma 4.2(d) it gives

$$\Sigma_{r,D}(x) = 16\, c_F\, \mathfrak{L} \sqrt{\frac{x}{rD}} + O\big(x^{1/4+\varepsilon} + x^{1/4+\theta/2+\varepsilon} R^{1-2\theta}\big), \qquad \mathfrak{L} \gg_\varepsilon R^{-\varepsilon}.$$

The main term is $\gg x^{1/2} R^{-1-\varepsilon}$, and the error terms are smaller by a factor $\ll x^{-1/4+2\varepsilon} R^{1+\varepsilon} + x^{-1/4+\theta/2+2\varepsilon} R^{2-2\theta+\varepsilon}$, which tends to 0 by the choice of ε. Hence $\Sigma_{r,D}(x) \gg x^{1/2} R^{-1-\varepsilon}$ for $(r, D) \in \mathcal{D}$.

Next, $\#\mathcal{D} \gg R^2 (\log R)^{-2}$. Indeed, there are $\gg R/\log R$ primes $r \equiv 1 \pmod 8$ in $(R, 2R]$. Fix one. The conditions $D \equiv D_a \pmod 8$ and $\left(\frac{3}{D}\right) = 1$ are satisfied by the primes in one residue class modulo 24: if $D_a \equiv 1 \pmod 4$ we take $D \equiv 1 \pmod 3$, so that $D \equiv 1 \pmod{12}$, and if $D_a \equiv 3 \pmod 4$ we take $D \equiv 2 \pmod 3$, so that $D \equiv 11 \pmod{12}$. By the prime number theorem in arithmetic progressions there are $\gg r/\log r$ such primes $D \in [ar, 2ar]$. Among them, at most one equals r, none is $\le 3a$ once $r > 3$, and the number with $H_{r,D}$ a square or three times a square is $O_\varepsilon(r^\varepsilon)$. Indeed, with $y = (D + 3ar)/2$, the equation $y^2 - s^2 = 3a^2r^2$ has at most $\tau(3a^2r^2)$ solutions with $y, s > 0$; and $y^2 - 3a^2r^2 = 3s^2$ forces $y = 3y_1$ and $s^2 - 3y_1^2 = -n$ with $n = a^2r^2$. If $g = \gcd(s, y_1)$, then $g^2 \mid n$, and $s/g \equiv \nu y_1/g \pmod{n/g^2}$ with $\nu^2 \equiv 3 \pmod{n/g^2}$; the solutions therefore fall into $\ll \sum_{g^2 \mid n} \#\{\nu \bmod n/g^2 :\ \nu^2 \equiv 3\ (n/g^2)\} \ll_\varepsilon r^\varepsilon$ orbits under multiplication by the units of norm 1 of $\mathbb{Z}[\sqrt3]$, and each orbit contains $O(\log r)$ solutions with $0 < y_1 \le ar$.

By Proposition 5.5,

$$C_a(x) \gg x^{-\varepsilon} \sum_{(r,D)\in\mathcal{D}} \Sigma_{r,D}(x) \gg x^{-\varepsilon} R^2 (\log R)^{-2} x^{1/2} R^{-1-\varepsilon} \gg x^{1/2+\varpi-3\varepsilon} \ge x^{1/2+\delta}.$$

This proves Theorem 1.1 for odd a. With the second error term of Proposition 4.3, which is $\ll x^{1/4+\theta/2+\varepsilon} R^{1-\theta}$ here, the same argument works for every $\varpi < \delta_\theta^{**}$ and proves Theorem 1.4 for odd a.

If $a = 2^k a'$ with a' odd, then $C_a(x) \ge C_{a'}(x/2^k)$: if $n'^3 + a'^3 \in \mathcal{S}$, then $(2^k n')^3 + a^3 = 2^{3k}(n'^3 + a'^3) \in \mathcal{S}$, because $2 \in \mathcal{S}$. This reduces the general case to odd a. □

6. Three-term patterns

Let $0 < a < b$. By [4, Lemma 2.1] we may assume that a or b is odd. We fix p_0, B and A as in [4, Lemma 2.3] and put $P = n + p_0$, $T = P + A$, $Q = 2P + B$ and $c = 2A - B$, so that $\{P, P + B, T\} = \{n, n + a, n + b\}$, $B > 0$ is odd and $A \notin \{0, B\}$. Let $v \ge 2$ and $d_0 \equiv 1 \pmod 4$ be as in [4, Proposition 3.2], and $J = v + 2$, $j = \lceil (v+2)/2 \rceil$ and $E = 2^{2j-v} \in \{4, 8\}$ as in [4, (4.1)]. For integers r, d let $M_{r,d} = (d + cr)^2 + 4A(B - A)r^2$, as in Lemma 2.2; thus $M_{1,d} = m_d$ in the notation of [4].

Lemma 6.1 (2-adic set-up)**.** *Let $r \equiv 1 \pmod{2^J}$, $d \equiv d_0 \pmod{2^J}$ and $u \equiv 0 \pmod{2^j}$. Then $v_2(u^2 + M_{r,d}) = v$ and $2^{-v}(u^2 + M_{r,d}) \equiv d \pmod 4$.*

Proof. We have $M_{r,d} - M_{1,d} = c(r-1)\big(2d + c(r+1)\big) + 4A(B-A)(r-1)(r+1) \equiv 0 \pmod{2^J}$. Hence $u^2 + M_{r,d} \equiv u^2 + m_d \pmod{2^J}$, and the assertion follows from [4, Proposition 3.2], since $J = v + 2$. □

Put $C_0 = |c| + 2\sqrt{|A(B-A)|} + B + 1$ and $C_1 = (2C_0 + |c|)^2 + 4|A(B-A)|$.

Definition 6.2. A pair (r, d) is *good* if r is a prime with $r \equiv 1 \pmod{2^J}$; d is a prime with $d \equiv d_0 \pmod{2^J}$; $C_0 r \le d \le 2C_0 r$; and $M_{r,d}$ is neither a square nor three times a square.

For a good pair put $L = rd$, $H = 2^{-v}M_{r,d}$, $T(\ell) = 2^{v-2}(E\ell^2 + H)/(rd)$ and

$$\Sigma_{r,d}(x) = \sum_{\substack{\ell\in\mathbb{Z}\\ rd\mid E\ell^2+H}} r\big(T(\ell)\big)\, F\Big(\frac{T(\ell)}{x}\Big).$$

With $u = 2^j\ell$ we have $2^v(E\ell^2 + H) = u^2 + M_{r,d}$, so $T(\ell) = (u^2 + M_{r,d})/(4rd)$.

Lemma 6.3. *For a good pair (r, d), the setting* (F) *holds with $E = 2^{2j-v}$, $H = 2^{-v}M_{r,d}$, $L = rd$, $\Lambda = \mathbb{Z}$, $s_0 = 0$ and $s = v - 2$, so that $\kappa_0 E = 2^{2j-2}$. Moreover $r^2 \le M_{r,d} \le C_1 r^2$.*

Proof. Since $d \ge C_0 r$ and $C_0 - |c| \ge 2\sqrt{|A(B-A)|}+1$, we have $M_{r,d} \ge (d-|c|r)^2 - 4|A(B-A)|r^2 \ge r^2$; the upper bound is clear. Lemma 6.1 with $u = 0$ gives $v_2(M_{r,d}) = v$, so H is odd. Since $N = EH = 4^{j-v}M_{r,d}$, (F1) holds. For (F2): $p_1 = r$ and $p_2 = d$ are distinct odd primes, as $d \ge C_0 r > r$. Modulo r we have $M_{r,d} \equiv d^2 \not\equiv 0$, and modulo d we have $M_{r,d} \equiv (c^2 + 4A(B-A))r^2 = B^2r^2 \not\equiv 0$, since $d > B$ and $d \ne r$. As $2^v(E\nu^2 + H) = (2^j\nu)^2 + M_{r,d}$, the congruence $E\nu^2 + H \equiv 0$ is solvable modulo r and modulo d because $-d^2$ is a square modulo $r \equiv 1 \pmod 4$ and $-B^2r^2$ is a square modulo $d \equiv 1 \pmod 4$; so $-EH$ is a non-zero quadratic residue modulo r and d. (F3) holds with $\Lambda = \mathbb{Z}$. For (F4), let $rd \mid E\ell^2 + H$. Then $E\ell^2 + H$ is odd, and Lemma 6.1 with $u = 2^j\ell$ gives $E\ell^2 + H = 2^{-v}(u^2 + M_{r,d}) \equiv d \pmod 4$, so $(E\ell^2 + H)/(rd) \equiv r^{-1} \equiv 1 \pmod 4$. □

Proposition 6.4. *Let x be large and let $\mathcal{D}$ be a set of good pairs with $r \le x^{1/5}$ for all $(r, d) \in \mathcal{D}$. Then*

$$\sum_{(r,d)\in\mathcal{D}} \Sigma_{r,d}(x) \le S_{a,b}(x) \cdot \max_{n\le x}\ r(T)\, r\big(4P(P+B)\big) \ll_\varepsilon S_{a,b}(x)\, x^\varepsilon,$$

with P and T as above.

Proof. Let $(r, d) \in \mathcal{D}$ and let ℓ give a non-zero term. Put $u = 2^j\ell$ and $T = T(\ell)$, so that $4rdT = u^2 + M_{r,d}$, $x/2 \le T \le x$ and $T \in \mathcal{S}$. Put $P = T - A$ and $n = P - p_0$. As in [4, Proposition 4.2], $1 \le n \le x$ and $P, P + B \ge 1$ for large x, and $\{n, n + a, n + b\} = \{P, P + B, T\}$. By Lemma 2.2, $P, P + B \in \mathcal{S}$, so n is counted by $S_{a,b}(x)$.

Fix n; then $Q = 2P + B$ is fixed. Attach to (r, d, u) the Gaussian integer $z = (rQ - d) + iu$. By Lemma 2.2, $|z|^2 = r^2N_0$ with $N_0 = 4P(P + B)$, and $z \notin r\mathbb{Z}[i]$ since $r \nmid d$. From $u^2 \le 2rdQ \le 4C_0r^2Q$ and $rQ - d \ge rQ/2$ (for large x) we get $|\arg z| \le 4\sqrt{C_0/Q} \le 8\sqrt{C_0/x}$, which is at most $1/(16R^2)$ with $R = \lfloor x^{1/5}\rfloor$ for large x. By Lemma 2.3 there are at most $r(N_0)$ pairs (r, z), and (r, z) determines $d = rQ - \operatorname{Re} z$ and $u = \operatorname{Im} z$, hence ℓ. This gives the first inequality, and the second follows from $r(n) \ll_\varepsilon n^\varepsilon$. □

Proof of Theorems 1.2 and 1.4 for $S_{a,b}(x)$. Let $\delta < \varpi < \delta^*_\theta$ and ε be as in Section 5, put $R = x^\varpi$, and let $\mathcal{D}$ be the set of good pairs with $R < r \le 2R$. By Lemma 6.3, $L \asymp H \asymp R^2$, so $LH \le x^{1/2}$ and $H \le x^{3/4}L$ for large x, and Proposition 4.3 (with $\omega(L) = 2$) gives

$$\Sigma_{r,d}(x) = \frac{32\, c_F\, \mathfrak{L}}{2^{j-1}}\sqrt{\frac{x}{rd}} + O\big(x^{1/4+\varepsilon} + x^{1/4+\theta/2+\varepsilon}R^{1-2\theta}\big) \gg x^{1/2}R^{-1-\varepsilon},$$

exactly as in Section 5. There are $\gg R/\log R$ primes $r \equiv 1 \pmod{2^J}$ in $(R, 2R]$, and for each of them $\gg r/\log r$ primes $d \equiv d_0 \pmod{2^J}$ in $[C_0r, 2C_0r]$. For fixed r, the number of such d with $M_{r,d}$ a square is at most $\tau(4|A(B-A)|r^2)$, since $(d+cr-s)(d+cr+s) = -4A(B-A)r^2 \ne 0$; and the number with $M_{r,d} = 3s^2$ is $O_\varepsilon(r^{2\varepsilon})$, since, as in Section 5, the solutions of $(d + cr)^2 - 3s^2 = -4A(B - A)r^2$ fall into $\ll_\varepsilon r^\varepsilon$ orbits under multiplication by the units of norm 1 of $\mathbb{Z}[\sqrt{3}]$, each containing $O(\log r)$ solutions with $|d+cr| \le 3C_0r$. Hence $\#\mathcal{D} \gg R^2(\log R)^{-2}$, and Proposition 6.4 gives $S_{a,b}(x) \gg x^{1/2+\varpi-3\varepsilon} \ge x^{1/2+\delta}$. The second error term of Proposition 4.3 gives Theorem 1.4 in the same way. □

Corollary 1.3 is the case $\theta = 7/64$ (respectively $\theta = 0$) of Theorems 1.1 and 1.2: $\delta^*_{7/64} = \frac{50/64}{456/64} = \frac{25}{228}$ and $\delta^*_0 = \frac{1}{8}$.

7. Remarks

Remark 7.1 (The case $r = 1$). For $n^3 + a^3$ and $r = 1$, Lemma 2.1 becomes $4Dm = u^2 + (D + 3a)^2 - 12a^2$ with $u = 2\ell$, the cubic analogue of the families of [4]: the polynomial has level D and discriminant $\asymp D^2$. For a prime $D \equiv D_a \pmod 8$ with $\left(\frac{3}{D}\right) = 1$ and $D > 3a$ the proofs of Lemmas 5.2 and 5.4 apply with $r = 1$, so (F) holds with $L = D$ and $\omega(L) = 1$, apart from $O(\log x)$ primes $D \le x$ for which $H_{1,D}$ is a square or three times a square. Since $LH \asymp D^3$ and $H/L \asymp D$, Proposition 4.3 applies for $D \le x^{1/3-\varepsilon}$. Its main term $\asymp \mathfrak{L}\sqrt{x/D}$ dominates the error terms when $D \le x^{(1-2\theta)/(5-6\theta)-\varepsilon}$, and D is determined by $z = (2m - 3a - D) + 2i\ell$, so each n arises from at most $r(4G_a(m))$ pairs (D, ℓ). Summing over $D \asymp x^{(1-2\theta)/(5-6\theta)-\varepsilon}$ gives $C_a(x) \gg x^{1/2+\delta}$ for every $\delta < (1 - 2\theta)/(10 - 12\theta)$, the exponent of [4].

Remark 7.2 (Optimality within the method). For a family of level L and discriminant $\asymp H$, Proposition 4.3 needs $L^{3/4-\theta/2}H^{1/4-\theta/2} \le x^{1/4-\theta/2-\varepsilon}$, and the family contributes $\asymp \sqrt{x/L}$ values; this is how δ^*_θ arises for $L \asymp H$. Smaller discriminants are possible, but they do not help. For the cubic families, $4H_{r,D} = (D - \sigma ar)(D + \sigma' ar)$ with $\sigma = 2\sqrt{3} - 3$ and $\sigma' = 2\sqrt{3} + 3$. If $D > \sigma ar$ and $D - \sigma ar \asymp J$ with $1 \le J \le r$, then $H \asymp aJr$ while $L \asymp ar^2$, so $H/L \asymp J/r$ can be small. Allowing such D (with the 2-adic conditions adjusted), the families with $r \asymp R$ and $D - \sigma ar \asymp J$ number $\asymp RJ$ up to logarithms, have level $\asymp R^2$, satisfy the condition of Proposition 4.3 when $R^{3/2-\theta}(JR)^{1/4-\theta/2} \le x^{1/4-\theta/2-\varepsilon}$, and together give $\asymp \sqrt{x}\, J$ values. Maximising over $1 \le J \le R$ gives the exponent δ^*_θ again, attained at $J \asymp R$. The same holds for three-term patterns. So δ^*_θ is the limit of rescaled families combined with Theorem 3.1.

Remark 7.3 (Level-aware pair counts). As explained in [4, Remark 13.1], a bound $\mathcal{P} \ll N^{1/2+\eta} + N^{1+\eta}/d$ for the number of pairs of Heegner points at bounded distance that are compatible with the level would replace $H^{1/4}$ in (1.2) by about $H^{1/4}(H^{-1/4} + d^{-1/2})$. With that bound, the families of [4] (for which one also needs the improvement of the main-term error described there, namely the convexity bound for $L(s, \psi)$ in place of the Pólya–Vinogradov inequality in Lemma 4.4) and the rescaled families both give every $\delta < 1/6$ (apart from the exceptional factor), so the rescaling does not change the conditional limit $1/6$ of [4, Remark 13.1]; it recovers part of the saving that such a bound would give, unconditionally. Going beyond $1/6$ requires averaging over families.

Remark 7.4 (Correlations and effectivity). As in [4, Remark 13.7], the proofs give $\sum_{n \le x} r(n)r(n + a)r(n + b) \gg x^{1/2+\delta}$ and $\sum_{n \le x} r(n + a)\, r(n^2 - an + a^2) \gg x^{1/2+\delta}$ for every $\delta < \delta^*_\theta$. Siegel's theorem enters only through Lemma 4.2(d); the argument of [4, Remark 13.6] based on the Landau–Page theorem should make the results effective, after discarding the pairs whose character ψ is induced by the exceptional character, but we have not checked the details.

8. Numerical checks

The scripts are in the folder `numerics` that accompanies this note. They use Python 3 with `numpy` and `scipy`, and one C program; some of them reuse scripts of [4].

Identities and the 2-adic set-up. For every odd $a \le 23$ we chose 6 random good pairs (r, D) with $9 \le r < 300$, and for each of them 25 random odd ℓ with $rD \mid \ell^2 + H_{r,D}$ and $m(\ell) \lesssim 10^{12}$; in all 1800 cases $m(\ell)$ is an integer with $m(\ell) \equiv 2 \pmod 8$, as Lemma 5.2 predicts, the identity (2.1) holds, and $G_a(m)$ is a sum of two squares by direct factorisation. For each of the eight patterns $\{0, 1, 2\}$, $\{0, 1, 3\}$, $\{0, 2, 3\}$, $\{0, 1, 4\}$, $\{0, 3, 8\}$, $\{0, 5, 8\}$, $\{0, 4, 7\}$, $\{0, 6, 11\}$ we chose 5 good pairs (r, d) and 20 values of ℓ each; in all 800 cases $(E\ell^2 + H)/(rd) \equiv 1 \pmod 4$, the identities of

Lemma 2.2 hold, and P and $P+B$ are sums of two squares by factorisation. For Remark 5.1 we took $a \in \{3,7,11,15,19\}$, random primes $r \equiv 1 \pmod 4$ and D with $\left(\frac{3}{D}\right) = 1$, and all $\ell \equiv 0 \pmod 4$ with $rD \mid \ell^2 + H_{r,D}$ up to $200rD$: in all 23200 cases the odd part of $m(\ell)$ is $\equiv 3 \pmod 4$.

The main term. Table 2 compares the sharp sums $\sum r(T(\ell))$ over $\ell \in \Lambda$ with $rD \mid E\ell^2 + H$ and $T(\ell) \le x = 10^{13}$ (both signs of ℓ) with the main term of Proposition 4.3 for the sharp cut-off, that is, $64\, c_\Lambda \mathfrak{L} \sqrt{x/(\kappa_0 EL)}$ (c_F replaced by $\int_0^1 t^{-1/2}dt = 2$); this case is not covered by Proposition 4.3. The constant $\mathfrak{L}$ was evaluated from Lemma 4.2(c), with $L(1,\psi)$ computed from the rapidly convergent series used in [4]. For each a and each pattern we took the first good pair with $r \ge 40$ and with $r \ge 200$ (for $\{0,1,4\}$, where $2^J = 64$, the first two primes $r \equiv 1 \pmod{64}$). In every case each $T(\ell)$ had the 2-adic shape required by (F4), and in the three-term cases P and $P+B$ were always sums of two squares.

case	r	D or d	L	H	$\#\{\ell\}$	$\sum r(T)$	main term	ratio
$a=1$	41	73	2993	4561	231210	1956304	1951998	1.002
	233	241	56153	58033	53380	320000	316711	1.010
$a=3$	41	167	6847	26437	152866	858192	857309	1.001
	233	719	167527	516661	30904	241840	238748	1.013
$a=5$	41	229	9389	52009	130542	1187984	1188073	1.000
	233	1213	282629	1469641	23794	168928	175036	0.965
$a=7$	41	347	14227	117709	106048	1347840	1350994	0.998
	233	1667	388411	2777917	20296	165360	166203	0.995
$\{0,1,2\}$	97	761	73817	128929	23278	71296	72039	0.990
	257	1913	491641	834433	9020	40880	39580	1.033
$\{0,1,3\}$	97	577	55969	38209	26732	66832	66208	1.009
	257	1409	362113	231937	10512	36256	37219	0.974
$\{0,2,3\}$	97	1109	107573	20445	19282	70272	68828	1.021
	257	2837	729109	102941	7404	35208	34858	1.010
$\{0,1,4\}$	193	2969	573017	1054653	8352	22800	23678	0.963
	257	3929	1009753	1852477	6292	39024	36752	1.062

TABLE 2. Sharp r-weighted counts along rescaled families at $x = 10^{13}$ against the main term of Proposition 4.3. For n^3+a^3: $E=1$, $\Lambda = 2\mathbb{Z}+1$, $\kappa_0 = 1$; for patterns: the data of [4, Table 3].

Proposition 4.3 bounds the first error term by a relative amount $\ll H^{1/2}x^{-1/4+\varepsilon}$, which is not small at this height for the families with the largest H. For the two largest deviations in Table 2, $a=5$ with $r=233$ and $\{0,1,4\}$ with $r=257$, the ratios at $x=10^{14}$ are 0.995 and 1.016.

Type I sums at composite level. We computed the sums Ξ of (3.1), with $\psi_1(t) = \phi(2t-3)$, $\psi_2 = \phi$, $\phi(t) = \exp(-1/(1-t^2))$, $X = 10^6$, $\kappa \in \{1,3\}$ and $K = 2^{i/2}X \le 64X$, for the levels $d = L = 2993$ ($a=1$, $H=4561$) and $d = L = 6847$ ($a=3$, $H=26437$) with both $E=1$ and $E=4$, and for $d = L = 73817$ with the polynomial $8\ell^2 + 128929$ of the pattern $\{0,1,2\}$. In all cases $|\Xi| \le 0.0005\, X^{1/2}H^{1/4}$, the largest value of $|\Xi|$ was 3.9 against $X^{1/2} = 1000$, and for $K \ge 4X$ we always had $|\Xi| \le 1.02\, \mathfrak{M}^{1/2}$, where $\mathfrak{M}$ is the main term. This is consistent with Theorem 3.1 and with the observations in [4, §14]. The computations do not cover the whole range of K needed in Lemma 4.5: there Theorem 3.1 is applied with K up to Lk_+, while the length of the sum is $X \asymp \sqrt{Lx/(\kappa_0 E)}$, so K/X reaches about $\sqrt{2L} \approx 77$ for $L = 2993$ and about $4\sqrt{2L} \approx 1540$ for $L = 73817$. We only checked $K \le 64X$.

Multiplicity. For $a=1$, $x = 10^{10}$ and all 10 good pairs with $r \le 200$, every $m \le x$ produced by the families arises from exactly one family (up to $\ell \mapsto -\ell$); for $a=3$ and the 29 good pairs with $r \le 120$, among 38772 distinct values of m six arise from two families and none from more.

Exponents. Table 1 was computed in exact arithmetic.

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